Using Large Language Models to Support ERP Requirement Classification: An Empirical Study – Supplementary Materials

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1 No answer predictions

Table 1 presents a summary of requirement counts classified as "no answer".

Table 1: Counts and rates of requirements classified as "no answer".

Dataset	Platform	Model	Iteration	n	Rate (%)
A	Perplexity	Gemini 3.1 Pro Thinking	10	5	5.0
A	Perplexity	GPT 5.4 Thinking	6	4	4.0
A	Perplexity	GPT 5.4 Thinking	5	1	1.0
B	Claude	Haiku 4.5	2	18	17.1
B	Perplexity	GPT 5.4 Thinking	9	4	3.8
B	Perplexity	Gemini 3.1 Pro Thinking	5	3	2.9
B	Perplexity	GPT 5.4 Thinking	5	2	1.9
B	DeepSeek	DeepSeek-V3.2	1	1	1.0
B	GPT	GPT 5.4 Thinking	6	1	1.0
B	Perplexity	GPT 5.4 Thinking	3	1	1.0
B	Perplexity	GPT 5.4 Thinking	10	1	1.0

For brevity, configurations and iterations producing no "no answer" classifications were omitted.

2 Definitions of performance metrics

The results of classification from a single LLM configuration, a single run and from a particular dataset can be represented as the confusion matrix C:

$$C = \begin{pmatrix} n_{SS} & n_{SC} & n_{SN} \\ n_{CS} & n_{CC} & n_{CN} \\ 0 & 0 & 0 \end{pmatrix}, \tag{1}$$

where rows represent actual categories ("standard", "customization", and "no answer", respectively), columns represent predicted categories ("standard", "customization", and "no answer", respectively), and n denotes case counts. Since "no answer" categories did not occur in the actual data, the operational 2×3 matrix was augmented with the last row with zero counts. This did not add or relabel any observations but provided the 3×3 matrix to enable the calculation of multiclass MCC.

The multiclass MCC was derived from the R_K coefficient for K categories originally proposed in [1]:

$$\mathrm{MCC} = \frac{cs - \sum_{k=1}^{K} p_k t_k}{\sqrt{\left(s^2 - \sum_{k=1}^{K} p_k^2\right)\left(s^2 - \sum_{k=1}^{K} t_k^2\right)}}, \tag{2}$$

where c denotes the number of correct classifications:

$$c = \sum_{k=1}^{K} C_{kk} = \mathrm{tr}(C), \tag{3}$$

s denotes the total classification count:

$$s = \sum_{i=1}^{K} \sum_{j=1}^{K} C_{ij}, \tag{4}$$

t_k denotes the count in the actual class k:

$$t_k = \sum_{j=1}^{K} C_{kj}, \tag{5}$$

p_k denotes the count of k class predictions:

$$p_k = \sum_{i=1}^{K} C_{ik}. \tag{6}$$

In case of our C matrix, the MCC can be computed as:

$$\text{MCC} = \frac{cs - p_S t_S - p_C t_C - p_N t_N}{\sqrt{\left(s^2 - p_S^2 - p_C^2 - p_N^2\right)\left(s^2 - t_S^2 - t_C^2 - t_N^2\right)}}, \tag{7}$$

where:

$$\begin{aligned} c &= n_{SS} + n_{CC}, \\ s &= N = t_S + t_C, \\ t_S &= n_{SS} + n_{SC} + n_{SN}, \\ t_C &= n_{CS} + n_{CC} + n_{CN}, \\ t_N &= 0, \\ p_S &= n_{SS} + n_{CS}, \\ p_C &= n_{SC} + n_{CC}, \\ p_N &= n_{SN} + n_{CN}. \end{aligned} \tag{8}$$

The remaining metrics were calculated as follows:

Customization recall:

$$R_C = \frac{n_{CC}}{t_C} = \frac{n_{CC}}{n_{CS} + n_{CC} + n_{CN}}. \tag{9}$$

Standard recall:

$$R_S = \frac{n_{SS}}{t_S} = \frac{n_{SS}}{n_{SS} + n_{SC} + n_{SN}}. \tag{10}$$

False-standard rate:

$$\text{FSR} = \frac{n_{CS}}{t_C} = \frac{n_{CS}}{n_{CS} + n_{CC} + n_{CN}}. \tag{11}$$

False-customization rate:

$$\text{FCR} = \frac{n_{SC}}{t_S} = \frac{n_{SC}}{n_{SS} + n_{SC} + n_{SN}}. \tag{12}$$

Customization precision:

$$P_C = \frac{n_{CC}}{n_{SC} + n_{CC}}. \tag{13}$$

Standard precision:

$$P_S = \frac{n_{SS}}{n_{SS} + n_{CS}}. \tag{14}$$

Customization F1:

$$F1_C = \frac{2P_C R_C}{P_C + R_C} = \frac{2n_{CC}}{2n_{CC} + n_{SC} + n_{CS} + n_{CN}}. \tag{15}$$

Standard F1:

$$F1_S = \frac{2P_S R_S}{P_S + R_S} = \frac{2n_{SS}}{2n_{SS} + n_{CS} + n_{SC} + n_{SN}}. \tag{16}$$

Accuracy:

$$\text{Accuracy} = \frac{nss + ncc}{N}. \tag{17}$$

Macro-weighted recall (equivalent to accuracy):

$$R_{\text{MW}} = w_S R_S + w_C R_C = \frac{t_S}{N}\frac{n_{SS}}{t_S} + \frac{t_C}{N}\frac{n_{CC}}{t_C} = \frac{n_{SS} + n_{CC}}{N}. \tag{18}$$

Macro-weighted precision:

$$P_{\text{MW}} = w_S P_S + w_C P_C = \frac{t_S}{N}\frac{n_{SS}}{n_{SS} + n_{CS}} + \frac{t_C}{N}\frac{n_{CC}}{n_{SC} + n_{CC}}. \tag{19}$$

Macro-weighted F1:

$$F1_{\text{MW}} = w_S F1_S + w_C F1_C = \frac{t_S}{N}\frac{2n_{SS}}{2n_{SS} + n_{CS} + n_{SC} + n_{SN}} + \frac{t_C}{N}\frac{2n_{CC}}{2n_{CC} + n_{SC} + n_{CS} + n_{CN}}. \tag{20}$$

If a particular category was never predicted in a particular run, its precision and F1 were set to zero rather than omitted from the aggregated metrics. In weighted metrics, the weights were: $w_S = t_S/N$ and $w_C = t_C/N$ for standard and customization categories, respectively.

3 Sensitivity to threshold in the prompt's decision rules

Figure 1 illustrates the MCC sensitivity to the threshold in the prompt's decision rules. A threshold indicates the minimum value in the prompt's rule: "If a standard module covers $\geq x\%$ of the requirement $\rightarrow$ STANDARD". The value used for the main experimentation was set to 80%.

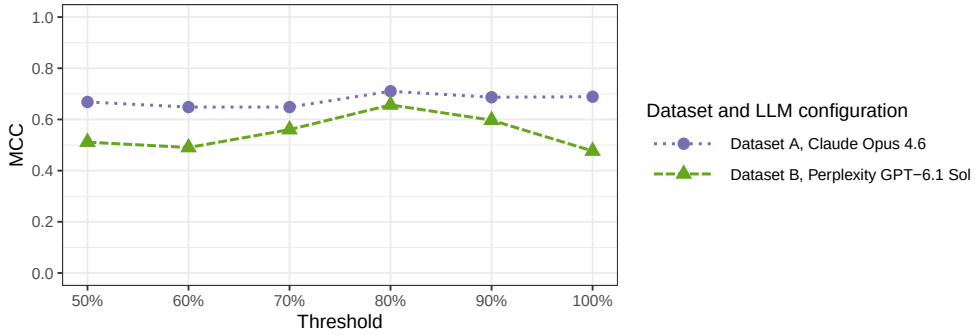


Figure 1: Sensitivity to threshold value in the prompt's decision rules.

4 Additional results for RQ1: Suitability of LLM configurations

Table 2: Overall performance metrics for individual LLM configurations in dataset A.

Rank	Platform	Model	MCC	MW F1	MW Precision	Accuracy
1	Claude	Opus 4.6	0.630 (0.051)	0.831 (0.022)	0.835 (0.024)	0.834 (0.022)
2	Claude	Sonnet 4.6 Thinking	0.620 (0.038)	0.825 (0.017)	0.832 (0.018)	0.831 (0.016)
3	Claude	Sonnet 4.6	0.618 (0.057)	0.825 (0.024)	0.830 (0.029)	0.829 (0.025)
4	Claude	Opus 4.6 Thinking	0.605 (0.040)	0.820 (0.018)	0.821 (0.018)	0.822 (0.019)
5	MS 365 Copilot	GPT 5.4	0.578 (0.056)	0.798 (0.028)	0.821 (0.031)	0.808 (0.026)
6	Gemini	Gemini 3.1 Pro	0.573 (0.044)	0.792 (0.020)	0.824 (0.023)	0.807 (0.016)
7	Perplexity	GPT 5.4 Thinking	0.563 (0.081)	0.795 (0.047)	0.808 (0.036)	0.792 (0.050)
8	GPT	GPT 5.4 Thinking	0.556 (0.036)	0.784 (0.023)	0.803 (0.015)	0.780 (0.025)
9	MiniMax	MiniMax-M2.7-Max	0.493 (0.064)	0.768 (0.031)	0.776 (0.028)	0.777 (0.027)
10	Z	GLM-5	0.446 (0.096)	0.720 (0.061)	0.789 (0.030)	0.757 (0.040)
11	MiniMax	MiniMax-M2.7-Air	0.438 (0.080)	0.724 (0.053)	0.775 (0.033)	0.754 (0.034)
12	Gemini	Gemini 3.1 Thinking	0.432 (0.061)	0.692 (0.040)	0.809 (0.014)	0.744 (0.025)
13	Z	GLM-5-Deep	0.413 (0.061)	0.688 (0.038)	0.794 (0.029)	0.739 (0.023)
14	DeepSeek	DeepSeek-V3.2	0.369 (0.079)	0.693 (0.029)	0.742 (0.044)	0.730 (0.027)
15	Claude	Haiku 4.5	0.344 (0.091)	0.688 (0.053)	0.723 (0.036)	0.719 (0.033)
16	Claude	Haiku 4.5 Thinking	0.282 (0.106)	0.650 (0.063)	0.712 (0.045)	0.701 (0.032)
17	Grok	Grok-4.20-Expert	0.263 (0.175)	0.626 (0.077)	0.713 (0.111)	0.700 (0.048)
18	Perplexity	Gemini 3.1 Pro Thinking	0.179 (0.036)	0.605 (0.035)	0.666 (0.045)	0.665 (0.013)

Values are means (SD) across ten iterations. MCC denotes the multiclass Matthews correlation coefficient. MW denotes the macro-weighted aggregation over two reference classes, "standard" and "customization". The reference class "no answer" received zero weight because such a category did not occur in the actual data. With this weighting scheme, accuracy equals macro-weighted recall. Rows sorted by Rank, according to the mean MCC.

Table 3: Overall performance metrics for individual LLM configurations in dataset B.

Rank	Platform	Model	MCC	MW F1	MW Precision	Accuracy
1	Perplexity	GPT 5.4 Thinking	0.589 (0.048)	0.817 (0.022)	0.821 (0.025)	0.816 (0.021)
2	GPT	GPT 5.4 Thinking	0.558 (0.059)	0.802 (0.026)	0.803 (0.025)	0.805 (0.025)
3	Claude	Sonnet 4.6 Thinking	0.524 (0.065)	0.775 (0.031)	0.803 (0.027)	0.793 (0.026)
4	Claude	Opus 4.6 Thinking	0.518 (0.085)	0.780 (0.040)	0.789 (0.035)	0.790 (0.034)
5	Gemini	Gemini 3.1 Pro	0.515 (0.106)	0.751 (0.066)	0.824 (0.026)	0.785 (0.044)
6	Claude	Sonnet 4.6	0.514 (0.045)	0.774 (0.023)	0.794 (0.019)	0.790 (0.018)
7	Claude	Opus 4.6	0.486 (0.064)	0.766 (0.028)	0.776 (0.030)	0.778 (0.025)
8	MS 365 Copilot	GPT 5.4	0.470 (0.098)	0.757 (0.046)	0.768 (0.041)	0.765 (0.041)
9	Perplexity	Gemini 3.1 Pro Thinking	0.443 (0.158)	0.722 (0.086)	0.797 (0.046)	0.762 (0.054)
10	MiniMax	MiniMax-M2.7-Max	0.432 (0.163)	0.722 (0.096)	0.784 (0.043)	0.759 (0.056)
11	Z	GLM-5	0.388 (0.086)	0.683 (0.051)	0.783 (0.033)	0.737 (0.031)
12	MiniMax	MiniMax-M2.7-Air	0.353 (0.077)	0.706 (0.038)	0.718 (0.032)	0.727 (0.026)
13	Claude	Haiku 4.5	0.347 (0.106)	0.693 (0.054)	0.727 (0.046)	0.714 (0.068)
14	DeepSeek	DeepSeek-V3.2	0.324 (0.108)	0.668 (0.065)	0.739 (0.038)	0.721 (0.035)
15	Z	GLM-5-Deep	0.320 (0.096)	0.638 (0.061)	0.787 (0.021)	0.713 (0.030)
16	Claude	Haiku 4.5 Thinking	0.303 (0.117)	0.662 (0.053)	0.721 (0.059)	0.714 (0.031)
17	Gemini	Gemini 3.1 Thinking	0.265 (0.083)	0.611 (0.051)	0.768 (0.030)	0.698 (0.025)
18	Grok	Grok-4.20-Expert	0.214 (0.112)	0.609 (0.054)	0.707 (0.077)	0.688 (0.028)

Values are means (SD) across ten iterations. MCC denotes the multiclass Matthews correlation coefficient. MW denotes the macro-weighted aggregation over two reference classes, "standard" and "customization". The reference class "no answer" received zero weight because such a category did not occur in the actual data. With this weighting scheme, accuracy equals macro-weighted recall. Rows sorted by Rank, according to the mean MCC.

Table 4: Classification performance for customization requirements in dataset A.

Rank	Platform	Model	Recall	Precision	F1	False-standard
1	Claude	Opus 4.6	0.697 (0.065)	0.808 (0.061)	0.745 (0.035)	0.303 (0.065)
2	Claude	Sonnet 4.6 Thinking	0.649 (0.041)	0.834 (0.041)	0.728 (0.028)	0.351 (0.041)
3	Claude	Sonnet 4.6	0.674 (0.034)	0.812 (0.074)	0.735 (0.031)	0.326 (0.034)
4	Claude	Opus 4.6 Thinking	0.714 (0.043)	0.765 (0.043)	0.737 (0.026)	0.286 (0.043)
5	MS 365 Copilot	GPT 5.4	0.609 (0.131)	0.830 (0.124)	0.683 (0.062)	0.391 (0.131)
6	Gemini	Gemini 3.1 Pro	0.540 (0.105)	0.876 (0.087)	0.658 (0.048)	0.460 (0.105)
7	Perplexity	GPT 5.4 Thinking	0.740 (0.068)	0.714 (0.104)	0.719 (0.044)	0.254 (0.067)
8	GPT	GPT 5.4 Thinking	0.811 (0.049)	0.654 (0.051)	0.721 (0.019)	0.189 (0.049)
9	MiniMax	MiniMax-M2.7-Max	0.563 (0.086)	0.744 (0.060)	0.636 (0.061)	0.437 (0.086)
10	Z	GLM-5	0.363 (0.131)	0.879 (0.073)	0.498 (0.136)	0.637 (0.131)
11	MiniMax	MiniMax-M2.7-Air	0.409 (0.155)	0.821 (0.107)	0.521 (0.126)	0.591 (0.155)
12	Gemini	Gemini 3.1 Thinking	0.277 (0.076)	0.976 (0.038)	0.426 (0.091)	0.723 (0.076)
13	Z	GLM-5-Deep	0.280 (0.082)	0.935 (0.087)	0.423 (0.089)	0.720 (0.082)
14	DeepSeek	DeepSeek-V3.2	0.326 (0.041)	0.777 (0.103)	0.458 (0.052)	0.674 (0.041)
15	Claude	Haiku 4.5	0.383 (0.175)	0.700 (0.089)	0.467 (0.136)	0.617 (0.175)
16	Claude	Haiku 4.5 Thinking	0.283 (0.181)	0.716 (0.127)	0.370 (0.167)	0.717 (0.181)
17	Grok	Grok-4.20-Expert	0.189 (0.145)	0.750 (0.274)	0.286 (0.174)	0.811 (0.145)
18	Perplexity	Gemini 3.1 Pro Thinking	0.197 (0.102)	0.633 (0.147)	0.278 (0.115)	0.794 (0.108)

Values are means (SD) across ten iterations. Recall is the ratio of correctly predicted customization requirements to the total actual customization requirements. Precision is the ratio of the correct customization predictions to the total customization predictions. False-standard rate is the ratio of actual customization requirements predicted as standard to the total actual customization requirements. Rows sorted by Rank, according to the mean MCC.

Table 5: Classification performance for customization requirements in dataset B.

Rank	Platform	Model	Recall	Precision	F1	False-standard
1	Perplexity	GPT 5.4 Thinking	0.692 (0.046)	0.773 (0.052)	0.729 (0.037)	0.308 (0.046)
2	GPT	GPT 5.4 Thinking	0.661 (0.070)	0.745 (0.037)	0.699 (0.046)	0.339 (0.070)
3	Claude	Sonnet 4.6 Thinking	0.489 (0.068)	0.843 (0.050)	0.616 (0.062)	0.511 (0.068)
4	Claude	Opus 4.6 Thinking	0.583 (0.120)	0.753 (0.061)	0.650 (0.079)	0.417 (0.120)
5	Gemini	Gemini 3.1 Pro	0.400 (0.146)	0.945 (0.051)	0.545 (0.146)	0.600 (0.146)
6	Claude	Sonnet 4.6	0.511 (0.068)	0.808 (0.049)	0.623 (0.049)	0.489 (0.068)
7	Claude	Opus 4.6	0.531 (0.066)	0.752 (0.059)	0.620 (0.052)	0.469 (0.066)
8	MS 365 Copilot	GPT 5.4	0.586 (0.151)	0.699 (0.089)	0.622 (0.095)	0.414 (0.151)
9	Perplexity	Gemini 3.1 Pro Thinking	0.361 (0.169)	0.892 (0.091)	0.490 (0.200)	0.639 (0.169)
10	MiniMax	MiniMax-M2.7-Max	0.422 (0.251)	0.819 (0.120)	0.501 (0.233)	0.578 (0.251)
11	Z	GLM-5	0.267 (0.107)	0.902 (0.086)	0.400 (0.119)	0.733 (0.107)
12	MiniMax	MiniMax-M2.7-Air	0.422 (0.114)	0.662 (0.063)	0.506 (0.092)	0.578 (0.114)
13	Claude	Haiku 4.5	0.383 (0.112)	0.708 (0.107)	0.484 (0.090)	0.608 (0.111)
14	DeepSeek	DeepSeek-V3.2	0.269 (0.144)	0.780 (0.097)	0.379 (0.160)	0.728 (0.146)
15	Z	GLM-5-Deep	0.186 (0.119)	0.952 (0.093)	0.291 (0.153)	0.814 (0.119)
16	Claude	Haiku 4.5 Thinking	0.258 (0.105)	0.737 (0.141)	0.371 (0.132)	0.742 (0.105)
17	Gemini	Gemini 3.1 Thinking	0.136 (0.088)	0.921 (0.094)	0.226 (0.126)	0.864 (0.088)
18	Grok	Grok-4.20-Expert	0.153 (0.107)	0.745 (0.211)	0.237 (0.139)	0.847 (0.107)

Values are means (SD) across ten iterations. Recall is the ratio of correctly predicted customization requirements to the total actual customization requirements. Precision is the ratio of the correct customization predictions to the total customization predictions. False-standard rate is the ratio of actual customization requirements predicted as standard to the total actual customization requirements. Rows sorted by Rank, according to the mean MCC.

Table 6: Classification performance for standard requirements in dataset A.

Rank	Platform	Model	Recall	Precision	F1	False-customization
1	Claude	Opus 4.6	0.908 (0.037)	0.849 (0.026)	0.877 (0.017)	0.092 (0.037)
2	Claude	Sonnet 4.6 Thinking	0.929 (0.021)	0.831 (0.015)	0.877 (0.012)	0.071 (0.021)
3	Claude	Sonnet 4.6	0.912 (0.040)	0.839 (0.014)	0.874 (0.021)	0.088 (0.040)
4	Claude	Opus 4.6 Thinking	0.880 (0.031)	0.852 (0.017)	0.865 (0.015)	0.120 (0.031)
5	MS 365 Copilot	GPT 5.4	0.915 (0.071)	0.817 (0.042)	0.861 (0.022)	0.085 (0.071)
6	Gemini	Gemini 3.1 Pro	0.951 (0.053)	0.796 (0.035)	0.865 (0.012)	0.049 (0.053)
7	Perplexity	GPT 5.4 Thinking	0.820 (0.097)	0.858 (0.024)	0.835 (0.052)	0.175 (0.100)
8	GPT	GPT 5.4 Thinking	0.763 (0.059)	0.884 (0.021)	0.817 (0.028)	0.237 (0.059)
9	MiniMax	MiniMax-M2.7-Max	0.892 (0.038)	0.793 (0.028)	0.839 (0.019)	0.108 (0.038)
10	Z	GLM-5	0.969 (0.019)	0.741 (0.038)	0.839 (0.022)	0.031 (0.019)
11	MiniMax	MiniMax-M2.7-Air	0.940 (0.047)	0.751 (0.045)	0.833 (0.019)	0.060 (0.047)
12	Gemini	Gemini 3.1 Thinking	0.995 (0.007)	0.719 (0.021)	0.835 (0.013)	0.005 (0.007)
13	Z	GLM-5-Deep	0.986 (0.020)	0.719 (0.021)	0.831 (0.012)	0.014 (0.020)
14	DeepSeek	DeepSeek-V3.2	0.948 (0.029)	0.723 (0.015)	0.820 (0.019)	0.052 (0.029)
15	Claude	Haiku 4.5	0.900 (0.069)	0.735 (0.048)	0.806 (0.021)	0.100 (0.069)
16	Claude	Haiku 4.5 Thinking	0.926 (0.060)	0.711 (0.047)	0.801 (0.014)	0.074 (0.060)
17	Grok	Grok-4.20-Expert	0.975 (0.017)	0.693 (0.041)	0.809 (0.025)	0.025 (0.017)
18	Perplexity	Gemini 3.1 Pro Thinking	0.917 (0.059)	0.683 (0.017)	0.782 (0.014)	0.080 (0.057)

Values are means (SD) across ten iterations. Recall is the ratio of correctly predicted standard requirements to the total actual standard requirements. Precision is the ratio of the correct standard predictions to the total standard predictions. False-customization rate is the ratio of actual standard requirements predicted as customization to the total actual standard requirements. Rows sorted by Rank, according to the mean MCC.

Table 7: Classification performance for standard requirements in dataset B.

Rank	Platform	Model	Recall	Precision	F1	False-customization
1	Perplexity	GPT 5.4 Thinking	0.881 (0.029)	0.846 (0.019)	0.863 (0.016)	0.107 (0.030)
2	GPT	GPT 5.4 Thinking	0.880 (0.023)	0.834 (0.028)	0.856 (0.017)	0.119 (0.023)
3	Claude	Sonnet 4.6 Thinking	0.952 (0.018)	0.782 (0.023)	0.858 (0.016)	0.048 (0.018)
4	Claude	Opus 4.6 Thinking	0.897 (0.042)	0.808 (0.042)	0.849 (0.023)	0.103 (0.042)
5	Gemini	Gemini 3.1 Pro	0.986 (0.015)	0.761 (0.042)	0.858 (0.024)	0.014 (0.015)
6	Claude	Sonnet 4.6	0.935 (0.024)	0.786 (0.021)	0.854 (0.011)	0.065 (0.024)
7	Claude	Opus 4.6	0.907 (0.027)	0.788 (0.023)	0.843 (0.017)	0.093 (0.027)
8	MS 365 Copilot	GPT 5.4	0.858 (0.078)	0.804 (0.048)	0.827 (0.033)	0.142 (0.078)
9	Perplexity	Gemini 3.1 Pro Thinking	0.971 (0.033)	0.747 (0.047)	0.844 (0.031)	0.025 (0.022)
10	MiniMax	MiniMax-M2.7-Max	0.935 (0.058)	0.766 (0.073)	0.837 (0.026)	0.065 (0.058)
11	Z	GLM-5	0.983 (0.016)	0.721 (0.029)	0.831 (0.016)	0.017 (0.016)
12	MiniMax	MiniMax-M2.7-Air	0.886 (0.041)	0.748 (0.031)	0.810 (0.015)	0.114 (0.041)
13	Claude	Haiku 4.5	0.887 (0.101)	0.737 (0.035)	0.803 (0.054)	0.091 (0.053)
14	DeepSeek	DeepSeek-V3.2	0.957 (0.028)	0.718 (0.037)	0.819 (0.017)	0.043 (0.028)
15	Z	GLM-5-Deep	0.988 (0.027)	0.701 (0.028)	0.820 (0.014)	0.012 (0.027)
16	Claude	Haiku 4.5 Thinking	0.952 (0.039)	0.712 (0.025)	0.814 (0.019)	0.048 (0.039)
17	Gemini	Gemini 3.1 Thinking	0.991 (0.010)	0.688 (0.021)	0.812 (0.012)	0.009 (0.010)
18	Grok	Grok-4.20-Expert	0.967 (0.035)	0.687 (0.024)	0.803 (0.016)	0.033 (0.035)

Values are means (SD) across ten iterations. Recall is the ratio of correctly predicted standard requirements to the total actual standard requirements. Precision is the ratio of the correct standard predictions to the total standard predictions. False-customization rate is the ratio of actual standard requirements predicted as customization to the total actual standard requirements. Rows sorted by Rank, according to the mean MCC.

Table 8: Confusion matrices for all LLM configurations in dataset A, pooled across all 10 iterations.

Rank	Platform	Model	Actual	Predicted		
				Standard	Customization	No answer
1	Claude	Opus 4.6	Standard	590 (90.8%)	60 (9.2%)	0 (0.0%)
			Customization	106 (30.3%)	244 (69.7%)	0 (0.0%)
2	Claude	Sonnet 4.6 Thinking	Standard	604 (92.9%)	46 (7.1%)	0 (0.0%)
			Customization	123 (35.1%)	227 (64.9%)	0 (0.0%)
3	Claude	Sonnet 4.6	Standard	593 (91.2%)	57 (8.8%)	0 (0.0%)
			Customization	114 (32.6%)	236 (67.4%)	0 (0.0%)
4	Claude	Opus 4.6 Thinking	Standard	572 (88.0%)	78 (12.0%)	0 (0.0%)
			Customization	100 (28.6%)	250 (71.4%)	0 (0.0%)
5	MS 365 Copilot	GPT 5.4	Standard	595 (91.5%)	55 (8.5%)	0 (0.0%)
			Customization	137 (39.1%)	213 (60.9%)	0 (0.0%)
6	Gemini	Gemini 3.1 Pro	Standard	618 (95.1%)	32 (4.9%)	0 (0.0%)
			Customization	161 (46.0%)	189 (54.0%)	0 (0.0%)
7	Perplexity	GPT 5.4 Thinking	Standard	533 (82.0%)	114 (17.5%)	3 (0.5%)
			Customization	89 (25.4%)	259 (74.0%)	2 (0.6%)
8	GPT	GPT 5.4 Thinking	Standard	496 (76.3%)	154 (23.7%)	0 (0.0%)
			Customization	66 (18.9%)	284 (81.1%)	0 (0.0%)
9	MiniMax	MiniMax-M2.7-Max	Standard	580 (89.2%)	70 (10.8%)	0 (0.0%)
			Customization	153 (43.7%)	197 (56.3%)	0 (0.0%)
10	Z	GLM-5	Standard	630 (96.9%)	20 (3.1%)	0 (0.0%)
			Customization	223 (63.7%)	127 (36.3%)	0 (0.0%)
11	MiniMax	MiniMax-M2.7-Air	Standard	611 (94.0%)	39 (6.0%)	0 (0.0%)
			Customization	207 (59.1%)	143 (40.9%)	0 (0.0%)
12	Gemini	Gemini 3.1 Thinking	Standard	647 (99.5%)	3 (0.5%)	0 (0.0%)
			Customization	253 (72.3%)	97 (27.7%)	0 (0.0%)
13	Z	GLM-5-Deep	Standard	641 (98.6%)	9 (1.4%)	0 (0.0%)
			Customization	252 (72.0%)	98 (28.0%)	0 (0.0%)
14	DeepSeek	DeepSeek-V3.2	Standard	616 (94.8%)	34 (5.2%)	0 (0.0%)
			Customization	236 (67.4%)	114 (32.6%)	0 (0.0%)
15	Claude	Haiku 4.5	Standard	585 (90.0%)	65 (10.0%)	0 (0.0%)
			Customization	216 (61.7%)	134 (38.3%)	0 (0.0%)
16	Claude	Haiku 4.5 Thinking	Standard	602 (92.6%)	48 (7.4%)	0 (0.0%)
			Customization	251 (71.7%)	99 (28.3%)	0 (0.0%)
17	Grok	Grok-4.20-Expert	Standard	634 (97.5%)	16 (2.5%)	0 (0.0%)
			Customization	284 (81.1%)	66 (18.9%)	0 (0.0%)
18	Perplexity	Gemini 3.1 Pro Thinking	Standard	596 (91.7%)	52 (8.0%)	2 (0.3%)
			Customization	278 (79.4%)	69 (19.7%)	3 (0.9%)

Values in brackets are row percentages, i.e., within the actual category. Rows sorted by Rank, according to the mean MCC.

Table 9: Confusion matrices for all LLM configurations in dataset B, pooled across all 10 iterations.

Rank	Platform	Model	Actual	Predicted		
				Standard	Customization	No answer
1	Perplexity	GPT 5.4 Thinking	Standard	608 (88.1%)	74 (10.7%)	8 (1.2%)
			Customization	111 (30.8%)	249 (69.2%)	0 (0.0%)
2	GPT	GPT 5.4 Thinking	Standard	607 (88.0%)	82 (11.9%)	1 (0.1%)
			Customization	122 (33.9%)	238 (66.1%)	0 (0.0%)
3	Claude	Sonnet 4.6 Thinking	Standard	657 (95.2%)	33 (4.8%)	0 (0.0%)
			Customization	184 (51.1%)	176 (48.9%)	0 (0.0%)
4	Claude	Opus 4.6 Thinking	Standard	619 (89.7%)	71 (10.3%)	0 (0.0%)
			Customization	150 (41.7%)	210 (58.3%)	0 (0.0%)
5	Gemini	Gemini 3.1 Pro	Standard	680 (98.6%)	10 (1.4%)	0 (0.0%)
			Customization	216 (60.0%)	144 (40.0%)	0 (0.0%)
6	Claude	Sonnet 4.6	Standard	645 (93.5%)	45 (6.5%)	0 (0.0%)
			Customization	176 (48.9%)	184 (51.1%)	0 (0.0%)
7	Claude	Opus 4.6	Standard	626 (90.7%)	64 (9.3%)	0 (0.0%)
			Customization	169 (46.9%)	191 (53.1%)	0 (0.0%)
8	MS 365 Copilot	GPT 5.4	Standard	592 (85.8%)	98 (14.2%)	0 (0.0%)
			Customization	149 (41.4%)	211 (58.6%)	0 (0.0%)
9	Perplexity	Gemini 3.1 Pro Thinking	Standard	670 (97.1%)	17 (2.5%)	3 (0.4%)
			Customization	230 (63.9%)	130 (36.1%)	0 (0.0%)
10	MiniMax	MiniMax-M2.7-Max	Standard	645 (93.5%)	45 (6.5%)	0 (0.0%)
			Customization	208 (57.8%)	152 (42.2%)	0 (0.0%)
11	Z	GLM-5	Standard	678 (98.3%)	12 (1.7%)	0 (0.0%)
			Customization	264 (73.3%)	96 (26.7%)	0 (0.0%)
12	MiniMax	MiniMax-M2.7-Air	Standard	611 (88.6%)	79 (11.4%)	0 (0.0%)
			Customization	208 (57.8%)	152 (42.2%)	0 (0.0%)
13	Claude	Haiku 4.5	Standard	612 (88.7%)	63 (9.1%)	15 (2.2%)
			Customization	219 (60.8%)	138 (38.3%)	3 (0.8%)
14	DeepSeek	DeepSeek-V3.2	Standard	660 (95.7%)	30 (4.3%)	0 (0.0%)
			Customization	262 (72.8%)	97 (26.9%)	1 (0.3%)
15	Z	GLM-5-Deep	Standard	682 (98.8%)	8 (1.2%)	0 (0.0%)
			Customization	293 (81.4%)	67 (18.6%)	0 (0.0%)
16	Claude	Haiku 4.5 Thinking	Standard	657 (95.2%)	33 (4.8%)	0 (0.0%)
			Customization	267 (74.2%)	93 (25.8%)	0 (0.0%)
17	Gemini	Gemini 3.1 Thinking	Standard	684 (99.1%)	6 (0.9%)	0 (0.0%)
			Customization	311 (86.4%)	49 (13.6%)	0 (0.0%)
18	Grok	Grok-4.20-Expert	Standard	667 (96.7%)	23 (3.3%)	0 (0.0%)
			Customization	305 (84.7%)	55 (15.3%)	0 (0.0%)

Values in brackets are row percentages, i.e., within the actual category. Rows sorted by Rank, according to the mean MCC.

Table 10: Distributions of predicted categories for all LLM configurations in dataset A, pooled across all 10 iterations.

Rank	Platform	Model	Predicted		
			Standard	Customization	No answer
1	Claude	Opus 4.6	696 (69.6%)	304 (30.4%)	0 (0.0%)
2	Claude	Sonnet 4.6 Thinking	727 (72.7%)	273 (27.3%)	0 (0.0%)
3	Claude	Sonnet 4.6	707 (70.7%)	293 (29.3%)	0 (0.0%)
4	Claude	Opus 4.6 Thinking	672 (67.2%)	328 (32.8%)	0 (0.0%)
5	MS 365 Copilot	GPT 5.4	732 (73.2%)	268 (26.8%)	0 (0.0%)
6	Gemini	Gemini 3.1 Pro	779 (77.9%)	221 (22.1%)	0 (0.0%)
7	Perplexity	GPT 5.4 Thinking	622 (62.2%)	373 (37.3%)	5 (0.5%)
8	GPT	GPT 5.4 Thinking	562 (56.2%)	438 (43.8%)	0 (0.0%)
9	MiniMax	MiniMax-M2.7-Max	733 (73.3%)	267 (26.7%)	0 (0.0%)
10	Z	GLM-5	853 (85.3%)	147 (14.7%)	0 (0.0%)
11	MiniMax	MiniMax-M2.7-Air	818 (81.8%)	182 (18.2%)	0 (0.0%)
12	Gemini	Gemini 3.1 Thinking	900 (90.0%)	100 (10.0%)	0 (0.0%)
13	Z	GLM-5-Deep	893 (89.3%)	107 (10.7%)	0 (0.0%)
14	DeepSeek	DeepSeek-V3.2	852 (85.2%)	148 (14.8%)	0 (0.0%)
15	Claude	Haiku 4.5	801 (80.1%)	199 (19.9%)	0 (0.0%)
16	Claude	Haiku 4.5 Thinking	853 (85.3%)	147 (14.7%)	0 (0.0%)
17	Grok	Grok-4.20-Expert	918 (91.8%)	82 (8.2%)	0 (0.0%)
18	Perplexity	Gemini 3.1 Pro Thinking	874 (87.4%)	121 (12.1%)	5 (0.5%)

Rows sorted by Rank, according to the mean MCC.

Table 11: Distributions of predicted categories for all LLM configurations in dataset B, pooled across all 10 iterations.

Rank	Platform	Model	Predicted		
			Standard	Customization	No answer
1	Perplexity	GPT 5.4 Thinking	719 (68.5%)	323 (30.8%)	8 (0.8%)
2	GPT	GPT 5.4 Thinking	729 (69.4%)	320 (30.5%)	1 (0.1%)
3	Claude	Sonnet 4.6 Thinking	841 (80.1%)	209 (19.9%)	0 (0.0%)
4	Claude	Opus 4.6 Thinking	769 (73.2%)	281 (26.8%)	0 (0.0%)
5	Gemini	Gemini 3.1 Pro	896 (85.3%)	154 (14.7%)	0 (0.0%)
6	Claude	Sonnet 4.6	821 (78.2%)	229 (21.8%)	0 (0.0%)
7	Claude	Opus 4.6	795 (75.7%)	255 (24.3%)	0 (0.0%)
8	MS 365 Copilot	GPT 5.4	741 (70.6%)	309 (29.4%)	0 (0.0%)
9	Perplexity	Gemini 3.1 Pro Thinking	900 (85.7%)	147 (14.0%)	3 (0.3%)
10	MiniMax	MiniMax-M2.7-Max	853 (81.2%)	197 (18.8%)	0 (0.0%)
11	Z	GLM-5	942 (89.7%)	108 (10.3%)	0 (0.0%)
12	MiniMax	MiniMax-M2.7-Air	819 (78.0%)	231 (22.0%)	0 (0.0%)
13	Claude	Haiku 4.5	831 (79.1%)	201 (19.1%)	18 (1.7%)
14	DeepSeek	DeepSeek-V3.2	922 (87.8%)	127 (12.1%)	1 (0.1%)
15	Z	GLM-5-Deep	975 (92.9%)	75 (7.1%)	0 (0.0%)
16	Claude	Haiku 4.5 Thinking	924 (88.0%)	126 (12.0%)	0 (0.0%)
17	Gemini	Gemini 3.1 Thinking	995 (94.8%)	55 (5.2%)	0 (0.0%)
18	Grok	Grok-4.20-Expert	972 (92.6%)	78 (7.4%)	0 (0.0%)

Rows sorted by Rank, according to the mean MCC.

5 Definitions of GLMMs used in RQ2

Table 12: Definitions of GLMMs used in RQ2.

Model	Analysis group	Data scope	Fixed effects	Random intercepts
H0	Req. heterogeneity	Full	dataset	platform_model + run_id
Hreq	Req. heterogeneity	Full	dataset	platform_model + run_id + req_id
H0+act	Req. heterogeneity	Full	dataset + actual_class	platform_model + run_id
Hreq+act	Req. heterogeneity	Full	dataset + actual_class	platform_model + run_id + req_id
R0	Req. factors	Full	dataset	platform_model + run_id + req_id
Ract	Req. factors	Full	dataset + actual_class	platform_model + run_id + req_id
Rlen	Req. factors	Full	dataset + length_chars_z	platform_model + run_id + req_id
R0(c)	Req. factors	Common	dataset	platform_model + run_id + req_id
Rarea(c)	Req. factors	Common	dataset + area	platform_model + run_id + req_id
Ract(c)	Req. factors	Common	dataset + actual_class	platform_model + run_id + req_id
Ract+len	Req. factors	Full	dataset + actual_class + length_chars_z	platform_model + run_id + req_id
Ract+area(c)	Req. factors	Common	dataset + actual_class + area	platform_model + run_id + req_id
Ract*len	Req. factors	Full	dataset + actual_class × length_chars_z	platform_model + run_id + req_id

All models are binomial GLMMs with logit link.
Models were fitted using maximum likelihood with Laplace approximation via lme4::glmer in R.
The suffix '(c)' denotes models fitted on the common-area subset.
Data scope 'common' denotes a reduced dataset containing area categories that appear in both datasets A and B.
The variable length_chars_z denotes standardised requirement description length.

6 Additional results for RQ2

Table 13: Detailed comparison of paired GLMMs for the RQ2 (Part 1).

Family	Model 1				Model 2				
	Abbr.	AIC	R^2_m	R^2_c	Abbr.	AIC	R^2_m	R^2_c	MOR
Baseline	H0	40457.734	0.000	—	Hreq	21929.172	0.001	0.735	17.081
Baseline	H0+act	29840.095	0.345	0.365	Hreq+act	21779.597	0.371	0.715	6.265
Standalone	R0	21929.172	0.001	0.735	Ract	21779.597	0.371	0.715	6.265
Standalone	R0	21929.172	0.001	0.735	Rlen	21921.555	0.035	0.735	15.975
Standalone	R0(c)	18728.572	0.006	0.757	Rarea(c)	18693.720	0.188	0.760	13.700
Adjusted	Ract	21779.597	0.371	0.715	Ract+len	21781.436	0.372	0.715	6.266
Adjusted	Ract(c)	18603.207	0.384	0.739	Ract+area(c)	18603.741	0.407	0.741	6.633
Interaction	Ract+len	21781.436	0.372	0.715	Ract*len	21771.041	0.392	0.715	5.888

R^2_m and R^2_c marginal and conditional Nakagawa R^2 for particular models.
MOR is the median odds ratio for model 2 computed from the requirement-level random intercept.

Table 14: Detailed comparison of paired GLMMs for the RQ2 (Part 2).

Family	Model 1	Model 2	LR χ^2	Δdf	p	p_{adj}	ΔAIC	ΔR_c^2	ΔR_m^2
Baseline	H0	Hreq	18 530.562	1	<0.001	<0.001	18528.562	—	0.001
Baseline	H0+act	Hreq+act	8062.498	1	<0.001	<0.001	8060.498	0.350	0.026
Standalone	R0	Ract	151.575	1	<0.001	<0.001	149.575	−0.020	0.370
Standalone	R0	Rlen	9.616	1	0.002	0.002	7.616	0.000	0.033
Standalone	R0(c)	Rarea(c)	46.852	6	<0.001	<0.001	34.852	0.003	0.182
Adjusted	Ract	Ract+len	0.160	1	0.689	0.689	−1.840	0.000	0.000
Adjusted	Ract(c)	Ract+area(c)	11.466	6	0.075	0.150	−0.534	0.002	0.023
Interaction	Ract+len	Ract*len	12.395	1	<0.001	<0.001	10.395	−0.001	0.020

LR χ^2 and p-values come from likelihood-ratio tests of compared models.
Raw p-values were adjusted within each family using the Holm method.
ΔAIC = $\text{AIC}_{\text{Model1}}$ - $\text{AIC}_{\text{Model2}}$, so positive values favour Model 2.
ΔR_m^2 and ΔR_c^2 denote changes in ΔR_m^2 and ΔR_c^2 from Model 1 to Model 2.

References

[1] J. Gorodkin, "Comparing two K-category assignments by a K-category correlation coefficient," *Comput. Biol. Chem.*, vol. 28, no. 5-6, pp. 367–374, Dec. 2004. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S1476927104000799